

**System-on-a-Chip Integration in the Semiconductor Industry:
Industry Structure and Firm Strategies**

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Abstract

Ever increasing on-chip integration in the semiconductor industry, spurred by miniaturization, is at the point where large-scale electronic systems can be put on a single chip. The proliferation of such “systems-on-a-chip” has important repercussions for the structure of the electronics industry, and the strategies of electronics firms. In a detailed case study, we apply a theoretical framework developed in Somaya and Teece (2000) to understand how system-on-a-chip integration is altering the balance between integrated approaches, components trading, and the licensing of “design modules” in the semiconductor industry. Consistent with the framework, we observe a burgeoning market for licensed design modules in the industry, along with the primarily in-house design approach being pursued by large integrated firms. Important technical and institutional factors that are shaping industry structure, and the strategies being pursued by different types of firms are documented. Based on the framework, implications are drawn for the strategies of firms responding to technological change of the kind engendered by system-on-chip in the semiconductor industry. We extend the Somaya-Teece framework to include firm strategies that seek to influence the institutional environment in which they operate, and thus alter the balance between competing organizational modes.

Keywords: Semiconductor industry, modularity, transaction costs, innovation

System-on-a-Chip Integration in the Semiconductor Industry: Industry Structure and Firm Strategies

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1. Introduction

The semiconductor industry is undergoing yet another strategic revolution that is poised to transform existing business models. Ever-advancing miniaturization continues to increase the number of design elements that can be placed on an integrated circuit (IC), creating the possibility that large scale electronic systems can be built on a single chip. This “system-on-a-chip” (SOC) movement is engendering a marked shift in the approach towards creating, reusing, and licensing designs for ICs. In this increasingly complex world of system chips, a market for sub-chip design modules has begun to develop that could lead to new patterns for the division of labor in the industry.

Our analysis of this emerging phenomenon revolves around the functional design inventions that we call design modules (DMs). These design modules are combined in different ways to make marketable electronic end products, which typically include a very large number of these inventions. We analyze the SOC phenomenon by using a theoretical framework developed in Somaya and Teece (2000), which describes how such multi-invention products may

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be assembled using three generic alternative organizational modes. First, firms may utilize only internally developed technologies to build integrated products entirely in-house. Second, specialized component firms may market inventions in embodied component form to system integrators, who make the end products. Finally, firms may specialize in making only the inventions, which are then licensed and integrated into products by other firms.

We analyze the emerging shift in the semiconductor industry through a comparison of the costs and benefits of making electronic products using these alternative organizational modes – viz., internal organization, licensing markets, or component markets. We find that the SOC paradigm makes the licensing of semiconductor designs attractive, and that industry players are fostering institutional changes to further reduce the transaction costs entailed in licensing. At the same time, many large semiconductor firms are pursuing a more established integrated model based on in-house designs, and are competing for dominance with the licensing approach. But there are also complementarities between the two approaches, with the integrated firms being increasingly open to licensing external designs.

The rest of this paper is organized as follows. Section two describes the development of system-on-chip integration and design trading in the semiconductor industry. Section three analyzes this phenomenon in the context of the Somaya-Teece framework, and explores its implications for the organizational modes used in building electronic end products. Section four examines the strategic responses of various industry participants to this shift. Section five summarizes our analysis of system-on-chip in semiconductors, and discusses its implications for industry structure and firm strategy in multi-invention contexts generally. Section six concludes.

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2. System-on-Chip, and Design Trading in the Semiconductor Industry

In the semiconductor industry, the drive to exploit ever-increasing miniaturization has led to a new trend towards putting large scale electronic systems on a single chip. For over three decades, following the well-known "Moore's Law", the number of transistors on an integrated circuit (IC) has more or less doubled every 18 months.¹ In the early 1960s, digital ICs contained fewer than 50 transistors. According to the 1999 International Roadmap for Semiconductors (SIA, 1999), in 2001, the maximum number of transistors that can be fabricated on a single logic chip will reach 320 million, which is approximately 100 times the number of transistors on the original Pentium chip introduced in 1993.² The emerging system-on-a-chip (SOC)³ trend implies the exploitation of this increasing "silicon real estate" to combine several different functions on the same IC instead of using it to enhance the functionality of specialized components – e.g., combining a processor with memory on one IC instead of making a higher capacity memory IC and a more powerful processor.

Although Dataquest, a market research firm, defines a system-on-a-chip as an IC with more than 100,000 gates of logic and incorporating at least one processor and one memory function on the same chip,⁴ current SOC's are typically much larger and more complex than this minimal definition. In addition to memory and processors, other functions such as protocol converters, signal processors, and various input and output controllers are also being integrated in SOC's. While the semiconductor industry has witnessed various levels of on-chip functional

¹ Moore's law was articulated by Dr. Gordon Moore, one of the founders of Intel, in 1965.

² Continuing miniaturization has been the sine qua non of change in the semiconductor industry, but the buildup of potential seems to give rise to industry-wide restructuring only periodically. A wave of start-ups, including Intel, appeared between 1966 and 1972 to exploit the new integrated circuit ("IC") technology, and again a new wave, including LSI Logic, entered in the early 1980s to specialize in ASICs (or "application specific ICs").

³ System-on-a-Chip is also called system-LSI, system-ASIC, or system-on-silicon.

⁴ Peter Brown, "Cautious Approach To VSIA," Electronic News: September 23, 1996. Gates are the basic building blocks of semiconductor digital circuits that are made from one or more transistors.

integration in the past,⁵ the recent focus on SOC promises radical improvements in cost, portability, and functionality of electronic end-products.

System integration on an IC has several advantages over integration on a conventional printed circuit board (PCB). First, it increases operating speeds, since on-chip integration bypasses the congested data buses on PCBs and enables data transfer at faster clock speeds. A related advantage is lower power consumption, since lower voltages are required for each signal. Reduction in the size and complexity of end-use products resulting from fewer chips and minimal off-chip circuitry is a third major advantage of SOC. Fourth, the consolidation of functions on a single chip can lead to lower unit manufacturing costs relative to fabricating several separate chips.⁶ Fifth, the replacement of PCB interconnects with pathways on a single chip improves system reliability. These advantages are not only enabling downstream electronics manufacturers to improve the size, cost, and performance characteristics of existing end-products, but also opening up new applications where bulky PCBs are unusable.

The enthusiasm for SOC in the semiconductor industry is being driven by a shift in the IC market from performance-oriented, computer-centric end uses to other applications that include low-end computers, intelligent consumer products, and a plethora of communication devices. As long as the high-end PC industry was the primary driver of the IC business, semiconductor firms competed aggressively to be the first to market with cutting edge “performance components” that were able to earn a premium over current technology. But recently, price-sensitive households

⁵ System-level integration in semiconductors has a long history, but the systems in question have become more complex. As far back as 1969, for example, a company called Mostek produced the first single chip calculator, which integrated basic functions like addition, subtraction, multiplication, and division (Borror, 1988: 77). The current SOC movement reflects a qualitatively different level of integration however, holding out the possibility of putting highly complex, multi-function electronic circuits entirely on a single chip.

⁶ However, SOC is currently uneconomical for the *entire* system in many applications (despite the name system-on-a-chip), and certain specialty functions are often integrated into the system through separate ICs. Nonetheless, there are usually untapped cost advantages in increasing the level of on-chip functional integration, even if the *entire* system is not literally reduced to a single chip.

eager to access the Internet have assumed a larger share of the PC consumer base,⁷ and portable electronic devices have grown into an important emerging market segment.

SOC-based systems fit excellently into this emerging market scenario. National Semiconductor Corp., for example, has put a PC on a single chip, replacing most of the dozens of separate chips typically found in a PC. National's SOC-PC has somewhat poorer performance than a leading-edge microprocessor and separate chip set, but this is not a liability in the "internet appliance" market targeted by National, where cost and "footprint" (i.e., size) are expected to be significant differentiators.⁸ In addition, new applications exploiting the convergence of digital communication and computing have also spurred demand for SOCs. Cellular telephony and hand-held communication devices, where small size and low power consumption are significant differentiating factors, are leading examples of SOC use.⁹ Other emerging applications include cordless telephones, wireless local loop house-side boxes, internet set-top boxes, video game consoles, modems, and disk drives.

As these markets grow, SOC is steadily expanding its share of the semiconductor market. According to Dataquest (Table 1), SOC revenue in 1997, the first year the data was collected, was \$7.6 billion, but by 2002 it is forecast to grow to nearly \$40 billion, representing 11.8% of total semiconductor sales. Dataquest predicted in November 1999 that SOC would account for 40% of chip revenues by 2010.¹⁰

[Table 1 about here]

⁷ In 1996, the "sub-\$1,000" PC became the fastest growing segment of the PC industry, and PCs began penetrating into a broader range of U.S. households. Then, in 1999, low-end PCs began to be offered as incentives in exchange for subscriptions to Internet service providers or to enhanced services such as AOL.

⁸ Dean Takahashi, "National Semiconductor Readies Chip That Takes Over Most PC Functions," Wall Street Journal Interactive: July 15, 1999.

⁹ Texas Instruments, for example, has developed a digital cellular chip for Nokia that includes a microcontroller, a Digital Signal Processor (DSP), logic, and several memory and analog modules. [David Lammers, "CICC: SLI push elbows speed aside" Electronic Engineering Times Interactive: May 17, 1999.]

The SOC movement resonates with an important problem being faced by the semiconductor industry. While the complexity and density of ICs has increased rapidly in keeping with Moore's law, improvements in the productivity of IC designers have failed to keep up. This has resulted in a so-called design productivity gap in the industry (Figure 1), which has grown substantially through the 90s. One way out of this situation could be the brute-force application of more engineers to design work. However, IC designers are already highly paid and in short supply, and increasing the size of design teams poses additional problems due to rapidly rising team coordination costs. Semiconductor firms hope that the reuse of existing designs in SOC products will help bridge this productivity gap. We find, however, that large chip firms have run into some problems with their attempts to institute design reuse (see Section 4.1).

[Figure 1 about here]

Even if firm-level reuse of designs is established, many elements needed for an SOC may not be available in-house and will need to be built afresh, causing delays in new product introduction. Moreover, as SOCs become larger and more complex, it will become increasingly difficult for firms to remain competitive in all the functional design elements that are being integrated into SOCs. The solution to these problems appears to lie in the young and fast-growing market for design modules (DMs), known in the industry as "IP blocks," "design cores," or "virtual components".¹¹ According to Dataquest, trade in "macro" DMs grew rapidly from \$16 million in 1995 to \$140 million in 1998 (Table 2).¹² Using a broader definition of

¹⁰ "SOC market opportunities favor fewer players, says research firm" Electronic Buyers' News, November 11, 1999.

¹¹ The many names and acronyms used for design modules in the industry can lead to much confusion, which is compounded when they have different meanings in other contexts. For example, what is commonly referred to as IP in the industry (a circuit design that performs some function) isn't legally IP (intellectual property) at all. Similarly, virtual components are not components in the conventional sense, and are therefore inconsistent with the concept of components as used later in this paper. We therefore prefer our neutral term "design modules" (DM).

¹² See Gary Smith, "Confusion In The 'IP' Market," Electronic News: March 15, 1999, for a definition of "macro cells". Data were obtained through personal communication with Dataquest analysts.

design module that it adopted in 1998, Dataquest reported another large jump in the value of DM trade from \$306 million in 1998 to \$417 million in 1999 (Table 2).¹³ Most of these traded DMs are being offered by so-called “chip-less” firms, which sell no semiconductor product of their own and rely entirely on licensing revenue from their designs.

[Table 2 about here]

SOC integration and design trading in semiconductors may be seen as complementary developments. The market for design modules could be instrumental in the effective orchestration of the industry’s innovative resources in making SOC products. However, as we will explain in the next section, if DM trading is to occur on a large scale, institutional underpinnings such as industry-wide compatibility standards and market intermediaries are necessary. The growth in the value of traded DMs indicates that these structures are being put in place. Later in this paper, we examine how these changes have unfolded, and the ways in which various industry players have responded strategically to them.

3. Organizational Choices in SOC Design

The emerging SOC-based industry structure typifies the historical experimentation between integrated and ever more fragmented organizational modes of production in the electronics industry. Just as specialization in components proliferated in PCB-based electronic systems, the SOC era is showing signs of industry fragmentation driven by specialization in the disembodied semiconductor designs that are being licensed between firms. We analyze this shift

¹³ Personal communication with Dataquest. The new definition reads: “a semiconductor intellectual property block is a pre-designed function to be implemented in a semiconductor device. In some cases the functions are parameterisable, allowing a degree of customisation. These functions include physical library functions (analog or digital), basic blocks (such as counters and MUXs) and system level macros (also known as cores or virtual components) - including memory blocks. Specific IP elements may be supplied in soft, firm or hard forms and may be implemented in ASIC, ASSP or PLD devices.”

in organizational modes in the industry using a framework for such “multi-invention products” introduced in Somaya and Teece (2000). Based on the framework, we identify alternative organizational approaches to commercializing electronic end products, and analyze organizational choice as resulting from tradeoffs in transaction costs between these alternatives. This section concludes with a discussion of the industry initiatives that have sought to reduce transaction costs in the licensing of semiconductor designs.

3.1 The Semiconductor Industry in the PCB Paradigm

In PCB-based electronic systems, large firms like IBM, DEC, AT&T (Lucent), and Motorola have often been highly integrated, manufacturing not only the final systems in-house, but also the most important ICs involved. On the other hand, many component manufacturers also operate in the industry, specializing in certain categories of general-use ICs – microprocessors, memory, graphics, etc. – which are traded in component markets for assembly onto PCBs by other firms. As open architectures proliferated in electronic systems industries generally, and especially in computers through the 1980s, component specialists have increased their share of revenues and rents at the expense of the more integrated players. In response, integrated systems firms have tried hard to match component firms in component markets, while continuing to compete in end products. Thus, under the PCB paradigm, both “integrated modes” and “component modes” of organizing electronic production co-exist (Figure 2), though over time, the component modes have been gaining an upper hand.

[Figure 2 about here]

In the next sub-section (3.2), we describe how these organizational modes are two of three generic alternatives in commercializing products that combine a large number of inventions, so called “multi-invention products”. The third, a “licensing mode”, is mostly absent

under the PCB paradigm. This does not mean that there was no licensing between semiconductor firms in the many decades that PCB-based products have been commercialized. Licensing occurred through broad cross-licensing deals, sharing of process IP, or within strategic alliances. But these were typically passive, IP-only licenses that merely removed impediments to commercialization posed by blocking legal IP rights (Grindley and Teece, 1997), and did not involve any actual transfers of know-how or designs between firms.¹⁴

The PCB era in the semiconductor industry also witnessed two crucial historical developments that created the context conditions for an effective licensing market in designs. The first of these was the establishment around 1980 of silicon-based CMOS technology as a dominant design in semiconductor process technology.¹⁵ As long as many different process technologies were in use, standardization in SOC interfaces, an important precondition for DM licensing, would have been very difficult to achieve. Although other processes continue to be used for certain IC applications, CMOS process solutions are rapidly being developed for many of them (Figure 3).

[Figure 3 about here]

The second key development was the emergence of design software that was capable of characterizing the CMOS process limits of different chip plants. This enabled the de-linkage of design and manufacturing in CMOS ICs, as reflected in the emergence, beginning in the early 1980s, of a burgeoning “fabless” segment – semiconductor design firms that outsource chip fabrication to “foundry” manufacturers (Macher, Mowery and Hodges, 1998).¹⁶

¹⁴ The need to ensure access to such blocking IP rights led to very aggressive patenting practices among semiconductor firms (Hall and Ham, 1999).

¹⁵ Over time, the temperature stability of silicon and low heat generating properties of CMOS (“complementary metal oxide semiconductor”, a semiconductor device design that incorporates both p-channel and n-channel transistors) has made it the process of choice for all but a few specialized applications.

¹⁶ The authors suggest (pp. 134, endnote 35) that the “diffusion of MOS production technology facilitated the division of labor between device designers in fabless firms, who were able to operate within relatively stable design

3.2 Organizational Trade-offs in the Somaya-Teece (ST) Framework

We analyze the evolution of the semiconductor industry using a framework introduced in Somaya and Teece (2000) (henceforth, the Somaya-Teece framework), which compares alternative ways to combine multiple inventions to make end products. As mentioned earlier, three generic approaches to commercialization are suggested by the framework, respectively labeled *integrated*, *licensing*, and *component* modes.

One way to understand the framework is through the stages at which inventions may be organizationally combined with others as one progresses towards commercialization of the end product (Figure 4). First, all relevant inventions may be combined within the same firm, where they are used internally to make the end product, resulting in an integrated mode. The integrated mode subsumes merger and acquisition activity through which ownership of inventions is consolidated into a single firm, and they may also witness some passive licensing.¹⁷ Second, in the licensing mode, inventions actually reside in separate firms, but are combined through “active” licensing that entails the transfer of technologies between firms. Finally, firms may elect not to license their inventions, but embody them in traded components that are used by other firms to build the multi-invention product, resulting in a components mode.

[Figure 4 about here]

In the case of SOC integration, we define design modules (DMs) as the relevant “unit of invention.”¹⁸ Under the PCB paradigm, IC designs are typically used to make separate ICs that

rules, and foundries, who were able to incrementally improve their process technologies to accommodate a succession of new device designs.”

¹⁷ When passive (IP) licensing is used, the firms typically already possess internally developed versions of the licensed technology (sometimes even before the owners of the IP rights are known), and hence can be described as “integrated”.

¹⁸ Each design module may be covered by several different pieces of intellectual property, including multiple patents and copyrights. But it is design modules, as against patented inventions, that offer the possibility of reuse and licensing, and are therefore the appropriate units of analysis.

are then assembled with others on a PCB. Depending on whether the ICs are freely traded or not, a component mode or integrated mode prevailed in the industry. With SOC, componentization of design modules is no longer technically feasible since the modules have to be integrated in pre-fabrication design, as well as in manufacturing (literally, since they are fabricated on the same wafer). The organizational choices available to the industry are thus reduced to integrated or licensing modes.¹⁹

According to the Somaya-Teece framework, the organizational modes likely to predominate in any given market segment can be determined by evaluating their relative organizational effectiveness, where effectiveness is measured by the advantages gained and costs incurred by organizing the associated transactions within firms or across firm boundaries. This “transaction cost” approach compares the costs and benefits of alternative organizational modes relative to a hypothetical ideal – that of well-functioning *frictionless* markets, which are treated as a datum. In the transaction cost literature, such markets are generally considered to be superior to bureaucratic internal organization within firms, but as transaction costs rise, internal organization becomes more attractive. Two types of transactions are of interest for our analysis – the licensing of inventions (design modules), and the sale of component products embodying the inventions. The framework suggests that when transaction costs in both licensing and components markets are high, an integrated mode will be favored. Likewise, when these transaction costs are low, either a licensing mode or a component mode is likely, depending on the relative level of transaction costs in each of these markets.

¹⁹ Multi-chip modules (MCMs), also called systems-in-a-package (SIPs), offer a different technique for accessing many of the benefits of SOC while retaining the option for component trade. Here, each chip is fabricated separately on silicon, but the separate ICs are integrated within a single chip package. MCMs have not yet become popular in the industry, but they may be used in emerging applications where size and power consumption is critical (e.g. in portable devices), and the SOC approach is impractical for technical or cost reasons. In this paper, we focus on system-on-chip alone because it is a significant recent development that is qualitatively different from a theoretical standpoint.

For example, early in the PCB era, high transaction costs associated with components sourcing led to internalization of the most important chip design and manufacturing stages, especially among leading mainframe computer firms. These transaction costs arose primarily from the systemic nature of the technology (Teece, 1996), which required substantial coordination of innovation, and the associated problems of acquiring capabilities or components from relatively underdeveloped external markets in real-time, so called “dynamic transaction costs” (Langlois, 1991; Langlois and Robertson, 1995).²⁰ By contrast, the development of the electronic systems of the time could be coordinated quite effectively through intra-firm teams. As these products became more and more complex however, the bureaucratic costs of such coordination became quite onerous, especially when compared to the modular strategies being pursued by entrants.

Modularization, spurred by increasing standardization of PCB interfaces, improvements in the capabilities of specialized semiconductor firms, and proliferation of open systems in the microcomputer era (such as those for the PC and client-server architectures), helped reduce transaction costs in components markets. Soon, component modes were widely adopted because they had strong advantages stemming from the strength of decentralized innovation in modular industrial systems (Langlois, 1992a; Langlois and Robertson, 1992). IC component markets offered the benefits of high-powered market incentives, with the best-in-class IC suppliers being rewarded with large market shares. Integrated manufacturers who continued to employ inefficient internal divisions that were not competitive with market performance did so at their own peril.²¹

²⁰ The transaction costs in this context are more difficult to characterize in terms of Williamsonian asset specificity (Williamson, 1985); alternative descriptions such as those used here, appear to point more directly to their source.

²¹ During this phase, large firms like IBM and DEC that continued to rely on integrated structures, saw erosion of their market positions, despite large installed bases. DEC has since ceased to exist as an independent company, and

While licensing modes provide similar benefits from modular innovation, they also entail many transaction costs, which we describe below. Comparatively, the transaction costs in components markets are low, reflecting the conventional coordination costs of exchanging goods between suppliers and integrators. These include the problems of ensuring quality and delivery from closely-tied suppliers, and the difficulties with coordinating design changes across a market interface, which are not too cumbersome when compared with licensing. Due to the relative effectiveness of component markets, the “active” licensing of design elements was virtually absent in PCB-based systems.

3.3 The Semiconductor Industry in the SOC Paradigm

Despite its advantages with PCB-based systems, with SOC, the component mode becomes untenable. Not only does on-chip integration require the simultaneous fabrication of all hypothetical “components” (since they are now on the same chip), but it also entails non-trivial integration of the designs into a single system, so that the SOC can not be readily subdivided into physical components. It is important to remember that SOC ICs are also ultimately assembled on PCBs, but the critical change being witnessed is the migration of a substantial portion of the circuitry, and hence of the “system”, onto the SOC.

Lacking the possibility of a components mode, the emerging SOC-based semiconductor industry features a healthy competition between a networked industry structure based on the licensing of design modules and firms pursuing integrated modes using largely in-house designs (Figure 5). Which of these two structures will eventually dominate, and how this is likely to vary by industry segment, can be evaluated by comparing their associated organizational costs.

IBM revived its business in part by forcing its captive divisions, including semiconductors, to compete on component markets.

Moreover, these organizational costs are likely to change over time as initiatives to reduce the transaction costs in licensing that we discuss in Section 3.5 play out, and organizational capabilities in integrating outsourced DMs evolve.

[Figure 5 about here]

An integrated mode for SOC anticipates the design of entire ICs within a single firm that has all the requisite technology in-house. These technologies may have either been developed internally or obtained through the acquisition of innovative firms. However, other firms may own legal IP rights that can block the use of these technologies, making passive, IP-only licenses (typically, as a part of cross-licensing arrangements) necessary to use them. In general, the integrated mode is a workable model for SOC, and corresponds to the approach that many large semiconductor firms have initially taken. Integrated modes economize on the many transaction costs of (active) licensing markets for DMs discussed below. Integrated firms also typically own their own manufacturing facilities, which can provide further coordination advantages over licensing when there are complementarities between design and process features (Monteverde, 1995). In addition, they can use more flexible intra-firm design interfaces, thus avoiding the performance penalties imposed by inflexible and slow-moving industry-wide standards.

Despite these apparent advantages, the integrated mode has some critical drawbacks. Integrated firms may not have all the required technologies in-house to design certain SOC products. Even if in-house development is possible without infringing on the intellectual property rights of other firms, the duplicative reengineering (or inventing around) required may still be costly and time consuming. “Buying” the needed technologies by acquiring other firms is also somewhat costly, involving the use of highly imperfect merger markets. Most importantly, integrated modes sacrifice the benefits of modular innovation, viz. targeted market incentives for

excellence in each DM (as against the end product), and fertile experimentation in alternative technologies and approaches. Furthermore, the internal political status quo that typically facilitates coordination across design domains under the integrated model can become a liability as it may actually block the adoption of best-in-class design elements from outside.

In comparison, the licensing of DMs may appear to be an attractive alternative to integrated modes, but licensing markets are also plagued with many transaction costs. These contractual difficulties are aggravated by the fact that innovation in the industry is extremely fast paced and licensing delays cannot be easily countenanced. An industry group estimates that the time required to locate, evaluate and license a DM is currently on the order of five months.²² By contrast, the time between introductions of new models of personal computers, an indicator of the pace of innovation in electronic systems, has shrunk in recent years to three months (Curry and Kenney, 1999).

3.4 Transaction Costs in SOC Licensing Markets

Licensing markets in DMs are plagued by four types of interrelated transaction cost problems described in Somaya and Teece (2000) – respectively, *technological interconnectedness*, *diffuse entitlements*, *value allocation*, and *monitoring and metering*.²³ Each of these, in turn, are slowing down the adoption of DM trading in the industry and forcing market players to seek out solutions to mitigate them.

When implemented in the same SOC, DMs need to be adapted and integrated with each other. Typically, these needs cannot be entirely foreseen, and consequently such *technological*

²² Virtual Component Exchange presentation in Santa Clara, CA, July 27, 1999.

²³ Two other types of transaction cost, relating to the transfer of tacit know how and the strategic isolation of rents from complementary assets (Somaya and Teece, 2000), are less relevant in DM licensing. We have also excluded from our analysis here the ordinary market-making transaction costs of locating and matching buyers and sellers, which are rapidly being reduced by the internet and B2B marketplaces.

interconnectedness implies that licensing contracts are somewhat open-ended, making it ambiguous as to who is responsible for solving problems that arise in integration. Licensing problems due to technological interconnectedness are reflected in the high levels of warranties and after-sales engineering support that DM suppliers are required to provide, and in the frequent complaints heard from licensees about lack of “cooperation” from DM suppliers in solving their design problems. While “plug and play compatibility” standards for DMs are a stated industry goal, it is still some distance from realization. Moreover, industry-wide standardization may not be a panacea for all cases; where the technology is extremely fast moving, bureaucratic, politics-ridden public standards bodies may be too slow to coordinate the development of cutting edge design platforms.

A second source of transaction costs – *diffuse entitlements* – arises from the typically unclear allocation of IP rights for inventions in the industry. It is usually very difficult to identify beforehand all firms that own the IP rights that cover any given technology, leaving the parties exposed to substantial liabilities in the event of future litigation. This ambiguity arises not only because the technological domain is complex and rich in patents, but also because uncovering applicable IP rights is expensive and time consuming, especially when many relevant patents are yet to be issued. Further, patentees have incentives to strategically delay issue of their patents in order to hold up potential IP licensees for more rents. Consequently, the DM licensing arena has seen licensee demands for indemnity against IP litigation, which can be very expensive to provide, especially for small independent DM suppliers.²⁴

Given the large uncertainties in the commercial prospects of technologies, *valuation* differences between the parties are notorious for causing impediments in licensing transactions

between firms (Merges and Nelson, 1990; 1994). Moreover, this is compounded by the fact that in multi-invention products, any technology is by definition used jointly with other technologies, making it difficult to evaluate any single technology's contribution to the generated surplus. Legal uncertainties about the strength and scope of IP protection covering an invention further compound the valuation problem.

Finally, transaction costs also accrue from the *monitoring and metering* needs of licensing agreements, which exacerbate the contracting problems in each of the categories above. When DMs are licensed, it is very difficult to determine, in advance, appropriate measures for its use (of concern to the licensor) or its performance (of concern to the licensee). Once a technology is licensed, the licensor may lose control over the specific uses to which it is put. Although these may sometimes be limited by contract, the licensee invariably has some room to maneuver, which can cause unanticipated losses to the licensor. Monitoring and auditing problems also make it difficult to evaluate which ICs actually use the licensed DMs, and to determine appropriate license fees if licensees reverse engineer or modify the licensed design. Absent robust tools to verify the performance of licensed DMs, licensors may also be concerned about the "quality" of designs they are getting, and about "allocating blame" for failures down the road. Standards being developed in the industry seek to address some of these measurement issues by defining objective specifications for evaluating and testing DMs.

In addition, DM providers are experimenting with licensing different kinds of DMs to mitigate licensing costs. DMs come in three formats that are differentiated by their extent of

²⁴ Protection is being sought not only for design patents, but also for related process patents needed to fabricate the chips ["Legal designs," Electronics Times: March 30, 1998]. Although nascent insurance markets have begun to form in this area, critical market inefficiencies make IP insurance quite costly and onerous for licensors.

embodiment, in increasing order, as “soft”, “firm”, and “hard” DMs.²⁵ Soft modules and hard modules reflect the advantages and disadvantages associated with markets for the ideal-type “inventions” and “components” respectively. Hard modules permit the fastest implementation because they have been verified for performance, area, and timing, which eliminates some of the most time-consuming tasks of SOC design. Soft modules, by contrast, offer much greater flexibility for adaptation to a specific application, but they have only been verified for functionality and require lengthy additional design and verification work. Soft modules also entail higher licensing costs, due to the risk from reverse engineering, and problems associated with technological interconnectedness and diffuse entitlements.

Reflecting this tradeoff, Advanced RISC Machines (ARM), an experienced vendor of RISC-based processor modules, only makes hard versions of its DMs available in the early stages of the product cycle. As the design becomes more commonly used and experience with it accumulates, ARM starts making “softer” versions available and allowing licensees to adapt it more closely to their needs.

To summarize our analysis so far, in the SOC-centric chip industry of the future, both integrated and licensing modes seem feasible alternatives, but each brings different organizational costs. As we will see below (in section 4), integrated firms have been quite successful in leveraging their internal capabilities into making SOC, but other players in the industry, including many new firms, have taken on the challenge of making a licensing mode work. In part, this has found expression in vigorous attempts to reduce transaction costs in DM licensing through the creation of industry-wide standards for SOC design and reuse, and new institutional arrangements to facilitate DM trading.

²⁵ Hard DMs are provided as mask-level data that is specially adapted to the process technology of a particular manufacturer, such as a foundry, while less embodied soft DMs provide a more abstract description of architecture

3.5 Standardization and Licensing Institutions in SOC Design

The current institutional drive for SOC standards is led by a broad coalition of firms who came together in the Virtual Socket Interface Alliance (VSIA) to establish open, plug-and-play compatibility standards (so called “virtual sockets”) in semiconductor designs.²⁶ When established in September 1996 with 35 founding members, VSIA was largely comprised of leading EDA software firms, design module vendors, and several prominent Japanese electronics firms. But other firms soon joined in, and VSIA now includes virtually every significant player in the semiconductor industry. VSIA’s process of standardization, while still incomplete, promises to lower the costs of reusing and trading designs not only by reducing the technological interconnectedness between DMs, but also by setting up techniques for testing and measuring the performance of designs. By June 2000, VSIA had released its specifications for about a dozen standards, including on-chip buses, system level hardware description, and mixed signal ICs, and was engaged in efforts to encourage their use by participating firms.

Two other trade groups are attempting to further reduce transaction costs in DM licensing. RAPID (Reusable Application-Specific Intellectual Property Developers), was established at about the same time as VSIA by firms that develop and sell design modules in order to improve access to information about them. Among other initiatives, RAPID has developed a catalog standard for featuring commercially available DMs on the Internet.²⁷ Another group, Virtual Component Exchange (VCX), was created in October 1998 through the joint initiative of “Scottish Enterprise” (a Scottish economic development agency), and a few

and algorithm with no physical layout information. Firm DMs lie somewhere in between.

²⁶ See the VSIA website: www.vsi.org/about.htm. The VSI Alliance is chartered to define, develop, authorize, test and promote open standard specifications relating to data formats, test methodologies, and interfaces.

²⁷ See RAPID website: www.rapid.org

major players from VSIA.²⁸ VCX is addressing business and legal issues related to trade in design modules through standard contracts, monitoring systems, a matchmaking service, and customized arbitration services.

Other efforts at promoting a DM market are also underway. In May 2000, some major Japanese electronics firms formed a Japan-centric DM exchange called IPTC. Meanwhile, a plethora of private efforts have also mushroomed, including “EDA portals” such as SiliconX, SpinCircuit, and Toolwire, and licensing intermediaries such as HelloBrain.com and Yet2.com. So far, these ventures claim, at least publicly, to be working together on complementary aspects of creating a DM market. However, in March 2000, Design and Reuse, a B2B licensing firm with its own DM catalog, split away from the Silicon Integration Initiative (SI2) led industry group (which includes VCX) that was developing a common XML tagging schema for classifying DM catalogs.²⁹

Some firms have also developed technological solutions that, as suggested by researchers in other contexts (von Hippel, 1998), can help reduce transactions costs in licensing. Artisan Components, a provider of design libraries, has created a software tool that places a “watermark” on design modules, which can be used to overcome monitoring problems in tracking their use.³⁰ IP Valuation, a Silicon Valley start-up, markets an algorithm for valuing DMs based on the use to which it will be put and the structure of the deal (e.g. whether it is based on up-front fees or running royalties).³¹ A New Hampshire firm, Topdown Design Solutions, has generated a

²⁸ See VCX website: www.vcx.org. As of June 2000, VCX had thirty-six members, and was scheduled to launch its on-line trading site in September.

²⁹ Peter Clarke, “Design and Reuse exits IP catalog standards program,” *Electronic Engineering Times*: March 20, 2000. SI2 represents “industry-leading silicon systems and tool companies focused on improving productivity and reducing cost in creating and producing integrated silicon systems” (online at: www.si2.org).

³⁰ The company has also submitted the technology to VSIA as a possible future standard [Richard Goering, “IP99: Exec extols free-library model,” *Electronic Engineering Times Interactive*: March 25, 1999].

³¹ Richard Goering, “Startup looks to put price tags on IP cores,” *Electronic Engineering Times*: February 8, 1999.

“simulation model” that allows a soft “IP block” to be evaluated by potential customers without revealing the code in which it is written, thus reducing concerns about reverse engineering.³²

By reducing transaction costs, these innovations and institutions are collectively fostering the growth of the nascent market in DM licensing. In the next section, we look at the strategic responses of various actors in the industry to this emerging SOC scenario in the context of their historical capabilities and the organizational tradeoffs discussed above.

4. In Search of Strategy

Organizational modes in semiconductors are naturally chosen based on more than just the costs and benefits described in Section 3. In particular, pre-existing differences in capabilities and DM ownership have had an important impact on organizational choices. Based on these differences, the semiconductor industry has broadly split into two camps with respect to strategies for the SOC era. The first camp consists of large integrated firms that currently own the bulk of the industry’s DMs. These firms are maintaining their emphasis on designing and manufacturing internally, taking advantage of external capabilities, i.e. traded DMs, only when necessary. The second camp consists of several specialized players, including smaller design-only firms, customer systems firms, chip foundries, and EDA software suppliers, that are developing a relatively new, networked business model of SOC design and manufacture configured around licensing markets for DMs. However, the stark differences between the integrated and networked groups mainly represent starting positions reflecting their respective historical capabilities, and these differences appear to be narrowing over time.

The initial differences between the two camps and their subsequent reduction are evident in the history of VSIA. When VSIA was established in 1996, several large semiconductor firms,

³² Ann Steffora, “Core Evaluation Made Easy,” *Electronic News*: July 5, 1999.

led by IBM, Motorola, and LSI Logic, expressed reservations about its standardization efforts and continued to pursue their own internal SOC business plans.³³ Even though VSIA's proposals promised industry-wide benefits, integrated firms that were already able to achieve SOC designs with in-house technologies were hesitant to encourage competition in SOC by facilitating a market for DMs. But as VSIA's standardization efforts made headway, it drew in most integrated firms, including the early dissenters.³⁴ By August 1999, VSIA's membership included over 180 firms and institutions, reflecting a broad commitment to common standards from all significant players in the industry.

4.1 Integrated Firms

To be considered "integrated" by our definition, semiconductor firms need only *design* their system-on-a-chip products with little or no external licensing. However, in addition, many integrated firms also have their own in-house fabrication divisions. Integrated SOC firms usually fall into one (or both) of two categories: those with strong historic ties to electronic systems divisions in the same company, and those with extensive experience creating ASICs (application-specific ICs) or microcontrollers (microprocessors with a block of memory devoted to a customer-specific instruction set) for various end-use markets. Firms falling in the first category, some of which are also leading ASIC suppliers, include IBM, Lucent and Motorola in the U.S., NEC, Hitachi and Toshiba in Japan, and Infineon (formerly the semiconductor division of Siemens) and Philips in Europe. ASIC suppliers with no connection to internal systems

³³ Brown [note 4 supra]. This response on the part of large integrated firms is not unusual, and parallels the initial response of integrated firms in the automobile industry (Langlois and Robertson, 1995, Chapter 5) and Applied Materials, the most integrated firm in the semiconductor equipment industry (Langlois, 1992b), to interface standardization efforts undertaken by the Society of Automotive Engineers (SAE) and Semiconductor Equipment and Materials International (SEMI), respectively.

divisions include LSI Logic and Texas Instruments in the U.S., and STMicroelectronics in Europe. Important microcontroller vendors not included in one of the preceding categories are Motorola and Intel.

A subset of these firms, shown in Table 3, have long relied on pre-configured “standard cells” (the larger of which are roughly equivalent to design modules) that can be reused in multiple designs, suggesting that they were already well positioned for system-level integration. But even in many of these firms, little communication occurred between IC design teams serving different end-user markets. Consequently, a single function could be re-designed several times within the same company instead of being designed once and reused. However, integrated firms have now recognized the need to adjust their internal organization and design practices to take advantage of SOC opportunities.

Building on their internal culture of modular design, integrated ASIC suppliers are reinforcing their design reuse practices to bridge the design productivity gap. According to a Dataquest survey, the average proportion of an ASIC that was reused from previous designs jumped from 22 percent in 1997 to 33 percent in 1998.³⁵ Integrated firms are also addressing the tougher challenge of building reusable modules from their vast product portfolios, which were not originally designed for reuse. Firms such as NEC, Motorola, and Alcatel Microelectronics even created special task forces for this purpose.³⁶ However, the initiatives to unpack legacy

³⁴ In March 1999, VSIA also removed one of the last barriers to full participation of the major firms, such as IBM, by permitting members to specify which patents would be made available to other members as part of a standard, instead of operating under a presumption of contribution.

³⁵ Peter Clarke, “Early users of IP cores could gain an edge from design reuse,” *Electronic Engineering Times Interactive*: June 22, 1999.

³⁶ Alcatel Microelectronics established an internal “design factory” and tried to involve the original engineers in the redesign and documentation [Luke Collins, “Europeans Capitalize On Design's New World Order,” *Electronic Engineering Times Interactive*: March 13, 1998]. NEC assembled teams of engineers from different divisions to rapidly build up a portfolio of reusable modules [Anthony Cataldo, “Japan struggles with design reuse in SOC era,” *Electronic Engineering Times Interactive*: December 3, 1998]. And, in 1998, Motorola formed the Advanced Systems Technology Laboratory with an expected staff of about 300 engineers working on both, reclaiming legacy

DMs and make them reusable have often encountered difficulties. For example, the DSP group in Lucent Technologies reportedly opposed efforts to make its designs available to other Lucent ASIC groups because it feared that this would undermine the market for its own stand-alone products.³⁷ Such conflicts of interest arising out of the diffuse and somewhat long-term benefits of reuse are typical, requiring innovative organizational solutions from many integrated firms.³⁸

Integrated firms have also been struggling with transforming internal design practices to enable reuse of *new* DMs. This transformation not only requires otherwise independent design units to adhere to a common design methodology, but also necessitates a major cultural change in the work habits of design engineers. Designing for reuse can almost double the work required to design a DM, while the benefits of reuse are realized later and possibly in some other part of the company. This makes it very difficult to convince design engineers to put in the extra effort. Design engineers also appear to have a “cultural” bias against using DMs designed by other divisions because they fear unforeseen problems in SOC integration that they are not equipped to fix. LSI Logic and Texas Instruments, which were pioneers in design reuse, had a head start in dealing with many of these problems,³⁹ but other firms, like IBM and NEC, are catching up.⁴⁰

The emphasis on design reuse in integrated firms underscores the preference for in-house capabilities in their SOC strategy. When gaps in internally available DMs are perceived,

designs and supporting reusability of new designs [David Lammers, “Motorola sets up lab aimed at systems-on-chip,” *Electronic Engineering Times Interactive*: July 14, 1998].

³⁷ Ristelhueber, Robert, “Shaking up the Old Order,” *Electronic Business*: January 1998.

³⁸ For example, NEC faced problems convincing its business units to devote resources to developing DMs that would mostly benefit parts of the company outside their own divisional profit centers. Toshiba sought to overcome similar internal organizational resistance by adopting an internal licensing program that enabled business units to internalize the company-wide benefits. [Cataldo, note 36 *supra*]

³⁹ LSI Logic started a DM reuse program called “CoreWare” in the early 1990s. Texas Instruments also began early, and had made a company-wide database of about 100 DMs available to its engineers by 1998. The company also maintains a slim “IP rescue squad” of about a dozen engineers to help engineers deal with the problems of reusing DMs [David Lammers, “Design reuse rules at TI’s ASIC operation,” *Electronic Engineering Times Interactive*: December 3, 1998.].

integrated firms often prefer to augment their DM portfolio through acquisitions or joint ventures rather than turn to licensing. Intel, for example, entered into a joint design project with Analog Devices and acquired DSP Communications to bolster its capabilities in the area of digital signal processing (DSP).⁴¹ Lucent and Motorola are similarly collaborating in their StarCore joint venture to develop DSP design modules.⁴² And major ASIC suppliers like LSI Logic have been bolstering their RF and mixed signal DMs through acquisitions.⁴³ Prima facie, it appears that acquisitions and collaborations have provided a viable mechanism for integrated firms to remain effective in the SOC world by absorbing new technologies developed externally.

Integrated firms are also awakening to the value of the emerging market in DMs to address SOC opportunities. In most cases, this appears to be a one-way street, with integrated firms licensing external design modules for use in their own IC products but not licensing out. LSI Logic, for example, used the MIPS RISC processor DM in its IC design for Sony's highly successful PlayStation video game.⁴⁴ Over time, this trend appears to be intensifying; IBM expects that up to half its DMs will eventually come from outside vendors.⁴⁵

Integrated firms do not necessarily view the use of external DMs as a threat to their dominance of the SOC market. SOC solutions require significant expertise in application-specific system design, as well as a deep understanding of the interactions between DMs and other elements, including software – areas where integrated firms have much to offer. Some

⁴⁰ Russ Arensman, "In Silicon's Vise", *Electronic Business*: March 1999. The number of cores available in IBM's core library rose from 30 in 1996 to about 100 in 1999. NEC set up a 50-person group at its US subsidiary to help the company convert from custom ASIC design work to a more reuse-oriented approach.

⁴¹ David Lammers and Stephan Ohr, "Intel, ADI ally to develop DSP core," *Electronic Engineering Times*: February 8, 1999.

⁴² Brian Fuller and Martin Gold, "DSP rivals, Motorola and Lucent, become partners," *Electronic Engineering Times Interactive*: June 2, 1998.

⁴³ LSI Logic bought the optical read channel division of NeoMagic ["LSI Logic acquires design team from NeoMagic, *Electronic Buyers' News*, April 14, 2000].

⁴⁴ Clarke [note 35 supra].

integrated firms also see their competitive advantage in leveraging unique, high-value DMs that they own, and see no harm in licensing the more commodity portions of the chip design from external sources. Moreover, only integrators who have a good system-level understanding of the design and have contributed a significant fraction of the actual DMs used can provide chip-level assurances, which are currently valued highly by customers. Thus, according to the prevailing view, an in-licensing program is a useful complement to the integrated firms' primarily in-house strategic approach.

Integrated firms currently seem reluctant to license their DMs separately from their integration and fabrication services. However, in July 2000, Motorola decided to start an out-licensing program for its DMs,⁴⁶ which may presage an era of more robust licensing competition between integrated firms and other DM providers. The primary advantage of this strategy for integrated firms is the ability to leverage any given DM over multiple SOC opportunities, which an integrated firm may find difficult to pursue alone. Ultimately, integrated firms may have no choice in this matter – if the markets for DMs are successful and licensing becomes routine and low-cost, they may be forced to compete DM for DM, just as integrated firms were eventually obliged to compete in component markets for PCB-based products.

4.2 Networked Firms

Start-ups and other firms that do not have all the in-house capabilities of integrated firms have been experimenting with a licensing mode approach of the type discussed theoretically in Section 3. These firms execute SOC projects through a networked value chain involving some

⁴⁵ Will Wade, "Design reuse revolution still trying to get off the ground", Semiconductor Business News: March 15, 1999.

⁴⁶ Chris Edwards, "Motorola to step up licensing of its semiconductor IP cores," Electronics Times: July 3, 2000. Motorola was motivated by its keenness to retain erstwhile customers of its ICs, who are migrating to a SOC business model that involves DM providers and foundries.

combination of design firms, customer systems firms, electronic design automation (EDA) software firms, integrators, and contract manufacturers. This business model is still in its early stages, and depends critically on the growing standards and institutions described in Section 3.5. Moreover, it emerged during one of the worst downturns in the industry's history, and perhaps only after the current expansion (which began in 1999) plays out will we be able to evaluate its success.

4.2.1 Independent DM Providers

As the industry's attention turned to third-party DMs in the 1990s, many independent DM firms mushroomed, and over 200 were set up by 1998.⁴⁷ However, several of these firms focused on commodity DMs, such as those meeting new public standards (e.g. USB), and did not have defensible long-term appropriability positions. Faced with the many difficulties of making DM licensing work in a nascent market, many firms failed or were acquired, and their number quickly fell to less than 100.⁴⁸

In fact, only a very small number of start-up firms have actually been successful as DM providers. In 1999, three such firms, ARM (Advanced RISC Machines), MIPS and Rambus, together accounted for more than fifty percent of the DM licensing business.⁴⁹ A second group of firms, including inSilicon,⁵⁰ and EDA software firms Synopsys and Mentor, have built a successful "commodity DM warehouse" model based on licensing a broad range of relatively low-value DMs, many of which were acquired by buying out start-up firms.

⁴⁷ Peter Brown, "Changing IP Market Might Mean Trouble," Electronic News: September 7, 1998.

⁴⁸ Ann Steffora, "IP Cores at the Crossroads," Electronic News: March 29, 1999.

⁴⁹ Barbara Jorgensen, "Merchant IP market growing," Electronic Business: July 2000 (citing Dataquest figures).

⁵⁰ inSilicon, formerly the semiconductor IP division of Phoenix Technologies, has a large customer base, including Compaq, Fujitsu, Hewlett-Packard, Hitachi, Intel, LSI Logic, Lucent, Philips, Siemens, Sony, STMicroelectronics and Toshiba, and has been adding to its DM library through acquisitions of smaller firms. [Richard Goering and

ARM, established in 1990 by Apple Computer, VLSI Technology, and Britain's Acorn to develop “reduced instruction-set computing” (RISC) microprocessor designs, was one of the earliest entrants into DM licensing. With over thirty licensees and a strong presence in many application markets, it is arguably the most successful firm in this business.⁵¹ Two other processor DM companies, MIPS and DSP Group, are also considered successful today, but they took many years to develop their designs and customer relationships into a sustainable business. Rambus, a 1990 California start-up, has capitalized on Intel’s selection of its PC memory interface as the next-generation standard, and built up a substantial licensee base.⁵²

These successful start-up DM firms appear to have pursued a number of key strategies. First, they focused on design niches that have multiple disparate applications, where licensing modes have an advantage over integration due to the ability to access a broader market. Second, they have avoided DMs that rapidly lose value through easy duplication of their functionality, e.g. public standards-based one-time designs. Instead, they have preferred “architectural” modules, such as processors, which improve over successive generations. Third, these firms have implemented effective appropriability strategies to protect rents from their knowledge assets. Not only have they used legal intellectual property protection where possible, but also harnessed other mechanisms such as rapid time-to-market, complementary capabilities in sales and support, and network effects.⁵³ In particular, network effects resulting from the growth of compatible IC

Margaret Quan, “Phoenix, Sand merge to attain IP critical mass,” Electronic Engineering Times Interactive: September 18, 1998.]

⁵¹ In September 2000, the company had a market capitalization of over \$10 billion on half yearly (2000) revenues of \$65.4 Million.

⁵² Rambus’ position in the industry is highly controversial and the company is facing multiple lawsuits challenging its patents and practices.

⁵³ DMs can be legally protected by an overlapping set of patent, copyright, and semiconductor maskworks protection rights, but none of these are a good fit to the precise kind of “invention” involved in creating DMs. That is not to say they are irrelevant. ARM has acquired over 60 patents, and recently sued another firm, picoTurbo, over elements of the latter’s microprocessor design.

software and DM-specific engineering talent in system integration, as well as the DM's proven robustness in multiple applications, have been significant success factors.

Successful firms have also been diligent in providing adequate sales and support. In a business where technological interconnectedness is a pervasive problem, these firms have committed the resources to help diagnose and solve customer problems, even though the ratio of support to design effort can often be as high as three-to-one. DM providers with a commodity warehouse model have also been attentive to these needs, typically capitalizing on scale and scope economies in providing support for multiple DM offerings, which are not available to start-up DM providers. In addition, these commodity warehouse firms offer designs that are considered more reliable and proven, and provide credible warranties (i.e. someone to sue) in the event of unforeseen problems with the design. Hence, it's no surprise that they have managed to build a successful business where many a start-up has failed.

4.2.2 EDA Software Suppliers

Electronic design automation (EDA) software, which automates various stages of chip design, simulation, and verification, emerged as a separate industry segment in the 1980s. EDA firms also help engineers develop design methodologies, and have been in the vanguard of promoting design practices that would support a DM licensing market.⁵⁴

EDA firms have long provided libraries of pre-tested design elements (called "cells") for use with their tools. Typically, these cell libraries are tailored to the design rules of one or more semiconductor manufacturers. Library cells can be as small as a few dozen logic gates, and are the basic building blocks from which modules and entire chip designs are constructed. As we

⁵⁴ Michael Keating and Pierre Bricaud, respectively from Synopsys and Mentor Graphics, two leading EDA firms, literally "wrote the book" on design reuse (The Reuse Methodology Manual, Kluwer Academic Publishers, 1998).

have seen in the previous section, EDA firms like Mentor Graphics and Synopsys have been quick to make the jump from library provision to DM distribution, in part by acquiring other DM firms and licensing their design modules along with quality customer support. Mentor sold its cell libraries in early 2000 and is now only in the DM business, but Synopsys continues to license both.

The third major EDA firm, Cadence Design Systems, has adopted a very different strategy, acquiring no DMs for its own portfolio, but specializing in design integration services for SOC products using third-party and customer modules. The design services business appears to have been a success for Cadence, and in July 2000, the company spun out the design services division as a stand-alone company under the name Tality.

4.2.3 Foundries

Pure-play foundries, which first appeared in Asia in the late 1980s,⁵⁵ rely entirely on IC manufacturing contracts from other firms to keep their costly high-end manufacturing capacity in constant use. To attract and retain these customers, they have started acting as brokers for design modules and offering design services. Their key allies in this endeavor have been independent cell library providers that tune their products to the design rules of the sponsoring foundry. A notable deal between TSMC, a leading Taiwanese foundry, and Artisan Components of California in August 1998 offered “free IP”, for which Artisan forgoes the up-front licensing fee that is typical in the industry, and instead receives royalties from TSMC for each chip fabricated

⁵⁵ Foundry services have long been provided by integrated chip manufacturers when they have excess capacity, but dedicated foundries alleviated some of the contractual hazards inherent in this excess capacity model. Although they were initially well behind industry leaders in terms of manufacturing technologies, foundries have since closed the gap. In addition to leading-edge process technology, these firms have developed flexible manufacturing systems that are the envy of many vertically integrated producers.

using its design cells. The free IP model was quickly copied by TSMC's chief rival United Microelectronics (UMC), also of Taiwan.

Artisan's new strategy led many small cash-constrained cell library providers to exit the business, and even large firms like Mentor sold out to focus their energies elsewhere.

Independent DM providers initially looked askance at the "free IP" model, and questioned whether foundry relationships added as much value as their other licensing deals.⁵⁶

Nevertheless, most DM firms, including ARM, ultimately decided that they could not ignore the rapidly growing customer base of the foundries and signed deals to make foundry-specific versions of their modules available.

To complement this DM licensing activity, foundries have also set up design services businesses that Dataquest has dubbed "ASIC Lite" because they do not offer the full range of design services available from regular ASIC suppliers.⁵⁷ For the time being, the emphasis at the foundries is on doing what is necessary to keep capacity utilized, with other businesses developing into profit centers only in the distant future.⁵⁸ Nonetheless, the new foundry model has increased competitive pressure on integrated ASIC firms because it siphons off customers who might otherwise absorb their excess capacity or even buy their SOC chips.⁵⁹

4.2.4 Systems Firms:

Systems firms, the assemblers and marketers of electronics end products to consumers, are confronted with difficult choices in the SOC world. Complete SOC solutions offered by

⁵⁶ Peter Brown, "Changing IP Market Might Mean Trouble", Electronic News: September 7, 1998.

⁵⁷ Smith [note 12 supra].

⁵⁸ Mark Lapedus and Sandy Chen, "The Wafer Chase," Electronic Buyer's News Online: December 16, 1998.

⁵⁹ Gina Fraone, "Is the ASIC business model broken?," Electronic Business: January 1999.

integrated firms not only diminish the role of their system-level know-how and system design capabilities, but also limit their ability to effectively differentiate their products.

However, systems firms can also exploit opportunities emerging from the networked side of the SOC industry. The availability of design integration services from foundries and specialized firms like Tality, along with the availability of licensed DMs, allows systems firms to exercise more control over the SOC design process for their products. Moreover, many systems firms have long had their own IC design teams, which are being positioned to act as internal SOC integrators, using licensed-in design modules. For example in 1999, Quantum Corporation, the volume leader in hard disk drives (HDD) at the time, licensed a microcontroller module from NEC, a digital signal processor module from Texas Instruments, and a “read-channel” module from Lucent to create a single-chip solution for its drives.⁶⁰ The company’s strategy is to control the design to leverage its system-level understanding of the application, and dual-source the fabrication to maximize its leverage in price negotiations.

But not all system companies choose to tap into the networked business model in this way; some are content to fine-tune an application-specific reference design from an (integrated) SOC supplier.⁶¹ These firms are then free to focus their energies on the marketing end of the business – identifying consumer preferences, and harnessing emerging technologies to satisfy them. Moreover, reference designs can typically be adapted to meet a systems firm’s product differentiation needs by using proprietary software. For example, Cirrus Logic, a fabless firm that introduced the data storage industry’s first single-chip solution, allows its customers to port

⁶⁰ David Lammers, “Cost crunch creates push for single-chip drive,” EE Times: May 31, 1999. In October, 2000, Maxtor, a rival firm, announced that it was acquiring Quantum’s HDD business.

⁶¹ A reference design of this type is known as a “customizable ASSP” (application-specific standard product).

their own software to the SOC's microcontroller, providing Quantum's competitors with a somewhat different commercialization alternative to the one used by Quantum.⁶²

5. Discussion

The SOC movement in the semiconductor industry and the concomitant creation of licensing markets for DMs provide an interesting test case for applying the Somaya-Teece framework. Firms experiencing technological change of the type driving the SOC movement typically face two central strategic questions. First, they would like to know how their industry is likely to be transformed by such an environmental shift. How will industry structure change? Which sets of business models are likely to win out? And, which external developments are likely to alter the balance between these business models? Second, they would like to know how best to respond to the technological change. What products, markets, and technologies should they focus on, and what competencies or resources would they need? How can they reinforce these strategic positions? Should they consider acquisitions, licensing (both in and out), or divestment?

The framework, as applied here, clarifies the relevant factors needed to answer these questions for semiconductor firms in the SOC era. At the outset, it is important to clarify that SOC integration may not be attractive for all segments of the industry, or even for all components in a system. Technical hurdles in on-chip integration of certain components may impede the use of SOC solutions in some cases, and competition from PCB-based component modes is a significant factor in others. Where the cost and functional (size, power, and system speed) advantages offered by SOC are not significant, and fast-paced innovation requires the

⁶² Lammers [note 60 supra].

freedom to mix and match components, PCB-based component architectures are likely to continue.⁶³

In applications where SOC integration is used, the Somaya-Teece framework can be used to compare alternative organizational modes for commercialization, and to make reasonable assessments as to which ones are most likely to be more successful. To this end, in this paper, we have evaluated the costs and benefits of organizing DM-related transactions through licensing modes or integration within firms. Our analysis is summarized in Table 4. Transaction costs stemming from various sources discussed earlier are the primary drawbacks of using licensing modes. However, through modularization, licensing markets also provide more targeted economic incentives, facilitate greater innovative experimentation, and enable widespread use of best-in-class technologies in SOC commercialization.

[Table 4 about here]

Overall, even though integrated and licensing modes coexist in the industry, the distribution of costs and benefits suggests that integrated modes currently enjoy the upper hand. However, various factors may shift the balance over time. The success of standardization efforts in modularizing innovation in IC design and the growth of institutional arrangements (and technologies) that support DM trading are likely to lower transaction costs in licensing modes, allowing them to become more effective than governance-intensive integrated modes. Integrated modes, in turn, may benefit from other developments, such as the emergence of effective markets for mergers and acquisition. If new technologies can be successfully absorbed within integrated firms through such markets, some, but not all, of the natural advantages of licensing modes could be neutralized. Moreover, even if SOC standardization efforts succeed, in fast

⁶³ For example, high-end, short-cycle products, such as servers and Internet routers, continue to be integrated on PCBs because of time-to-market and technical performance considerations.

moving application domains, the public standards process may be too slow and bureaucratic to effectively respond to the needs of innovation. Consequently, integrated modes, with their internally governed proprietary interfaces, may be more effective at producing cutting edge performance SOC products.

In response to the second set of strategic questions we posed, firms in the semiconductor industry are clearly pursuing a number of different strategies to address the technological shift created by SOC. In this context, the Somaya-Teece framework is useful as a tool to figure out which mode – licensing or integrated – favors the capabilities of different firms. Historically integrated firms, for example, are currently viewing their high-quality DM libraries, system design capability, and connections to in-house fabrication as distinct advantages in pursuing an integrated approach to SOC commercialization. By contrast, smaller firms and start-ups, recognizing their limited capabilities, have sought to fill key roles in the emerging networked licensing mode, and, in some cases, have even extended themselves into allied activities where they have perceived opportunities.

Clearly, these strategies are contingent upon the current balance between the alternate organizational modes, where both licensing and integrated modes are feasible. If the context conditions for more effective licensing modes come about, integrated firms will need to become even more open to the licensing (in and out) of design modules, and loosen their ties to in-house DMs that do not measure up to market alternatives. Similarly, if ongoing efforts are unable to mitigate transaction costs in licensing markets, many network-side firms may be forced to rethink their focused strategies and consolidate into more integrated structures.

The framework also provides an insight into the strategic positioning of different firms within the SOC market. As we have seen, DM providers have been successful in those industry

segments where licensing modes offer the greatest advantages – e.g. in architectural designs that have many disparate applications. Similarly, if licensing modes becomes dominant in the future, integrated firms may respond by focusing on those market segments where integration offers significant benefits over licensing; e.g., where public standards are unresponsive to the needs of fast paced innovation.

SOC-related developments have also spurred firms into identifying lacunae and building new capabilities to fill them. Many large firms, for example, acquired or jointly developed new DMs to address gaps in their technology portfolio and leverage themselves into the integrated SOC business. Similarly, on the networked side, foundries perceived a need to develop DM licensing and integration operations to keep their capacity utilization high, and some systems firms have chosen to upgrade their in-house IC design capability for SOC. Consistent with the Somaya-Teece framework, networked firms have made a conscious decision to rely on external capabilities in many areas where markets to access them are effective.

The framework also suggests ways in which firms can influence their environment to their advantage. For example, the success of standardization efforts to make licensing more effective depends on both technical and organizational factors. While the technical feasibility of effective standards is primarily an exogenous factor, the industry-wide cooperation required in standards setting is open to strategic influence from industry participants. Firms may choose to encourage or impede these efforts depending on whether or not a licensing mode will benefit them. Indeed, we have seen that large integrated firms were not very helpful to VSIA in its early stages. Only when it became evident that there was considerable momentum behind VSIA did they join in to make sure that the new standards would not be detrimental to their interests.

Similarly, networked firms were in the vanguard of organizations such as VSIA, VCX, and RAPID since they had the most to gain from the success of licensing markets.

Moreover, once such market facilitating efforts gain momentum, individual firms and other new coalitions begin to address gaps in the evolving structure, thus creating positive spillovers for the entire process. For example, as the licensing of third-party DMs made inroads, new design portals, DM catalogs, and DM evaluation and encryption firms have mushroomed, which in turn has begun to further lower transaction costs in DM licensing markets. Thus, early strategic actions can have a disproportionate impact on eventual industry outcomes, and constitute an important endogenous fulcrum around which incumbents can leverage their historical position in the industry.

In sum, the SOC movement in the semiconductor industry has been expanding from the domain of integrated large firms to include greater numbers of networked firms in its ambit. Despite recent successes, it is not yet clear if, and to what extent, the licensing approach to SOC will be successful in the long run. Looking forward, the general framework we have introduced here can be used as an organizing template in multi-invention contexts like SOC to understand industry evolution, and as a guide to key strategic decisions in positioning, resource allocation, licensing, capability acquisition, and divestment.

6. Conclusion

In this essay, we have provided an analysis of the early-stage infrastructure and resulting firm strategies for SOC integration in the semiconductor industry. Based on our research in the trade press and interviews with a small number of experts, we have sought to systematically present a number of key facts about the SOC phenomenon, and inductively organize them

around an emerging theory of organizational choice in highly innovative contexts. We believe that the general approach of our analysis can be extended to other multi-invention industries, although our specific conclusions may have little external validity outside the SOC-centric electronics industry.

Overall, the theoretical premise that both integrated and licensing modes have their merits for SOC integration appears to have been borne out. While historically integrated firms began with an integrated approach to SOC, smaller design firms, foundries, customer systems firms, and EDA software producers appear to be successfully pioneering a networked model based on licensing markets. The success of ongoing institutional and infrastructure developments will determine the prominence of each of these organizational modes in the future.

To the existing literature on modularity and organizational choice (Langlois and Robertson, 1992, 1995; Sanchez and Mahoney, 1996), this paper has added the study of modularity in innovation, where organizational choices include a licensing mode, in addition to components and integrated ones. We have also extended the Somaya-Teece framework to include dynamics – in the endogenous relationship between firm behavior and evolving industry structure – and history – in the role of pre-existing capabilities when discontinuous change occurs. Lastly, although we have not focused on it in this paper, the framework could also be extended to address important questions of public policy.

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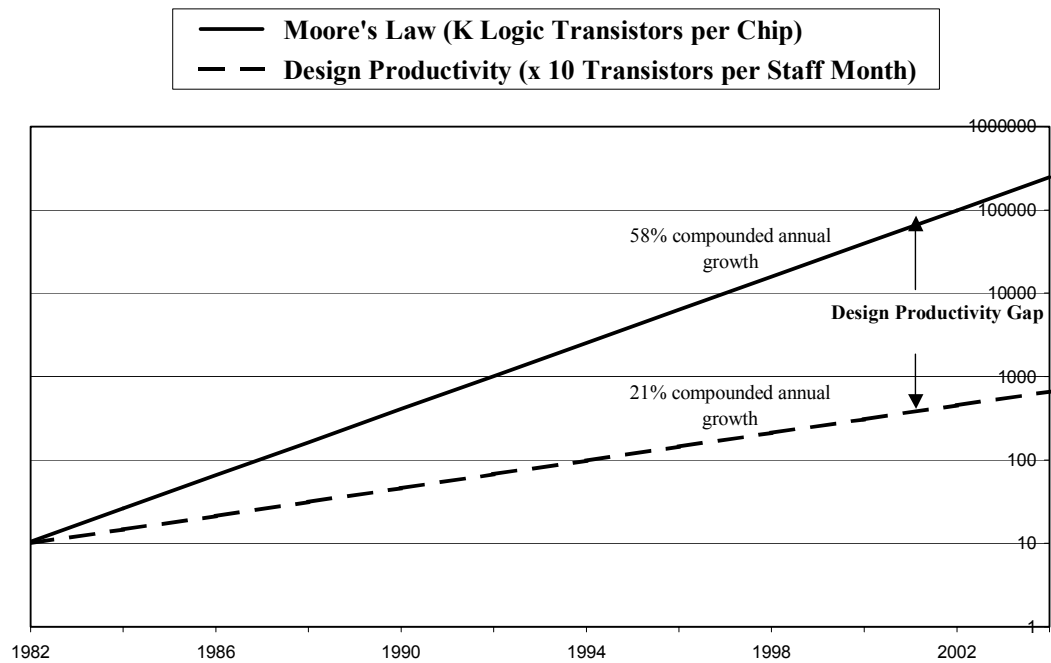


Figure 1: Widening Design Productivity Gap in Integrated Circuits
(Source: SIA, 1999)

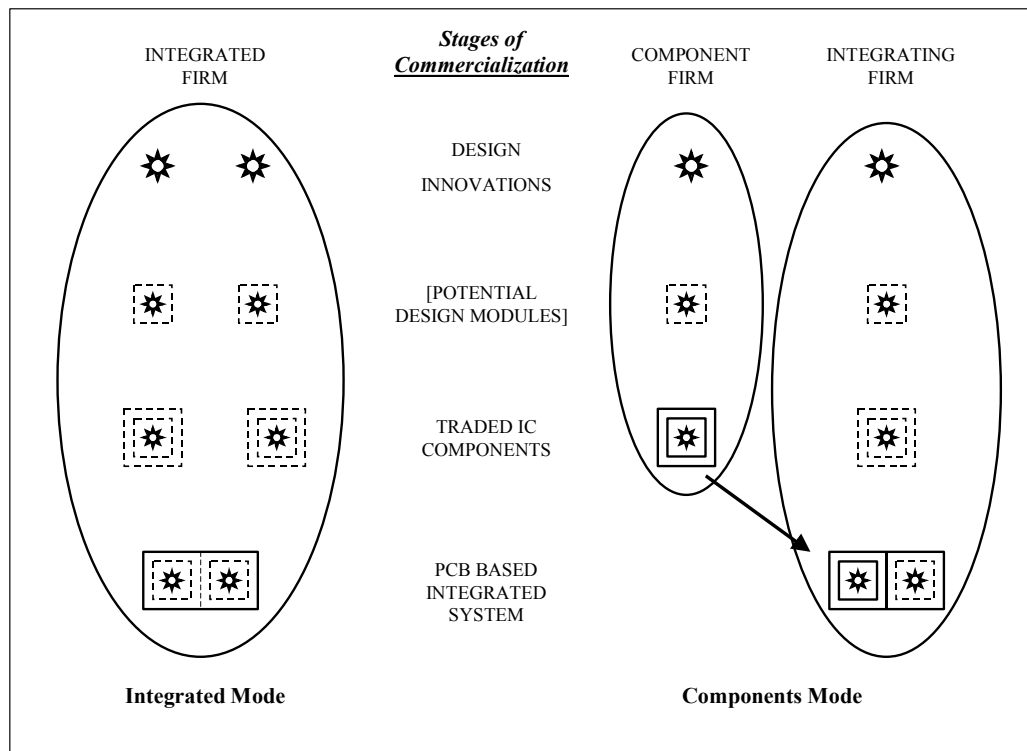


Figure 2: Integrated and Component Modes Under PCB Integration

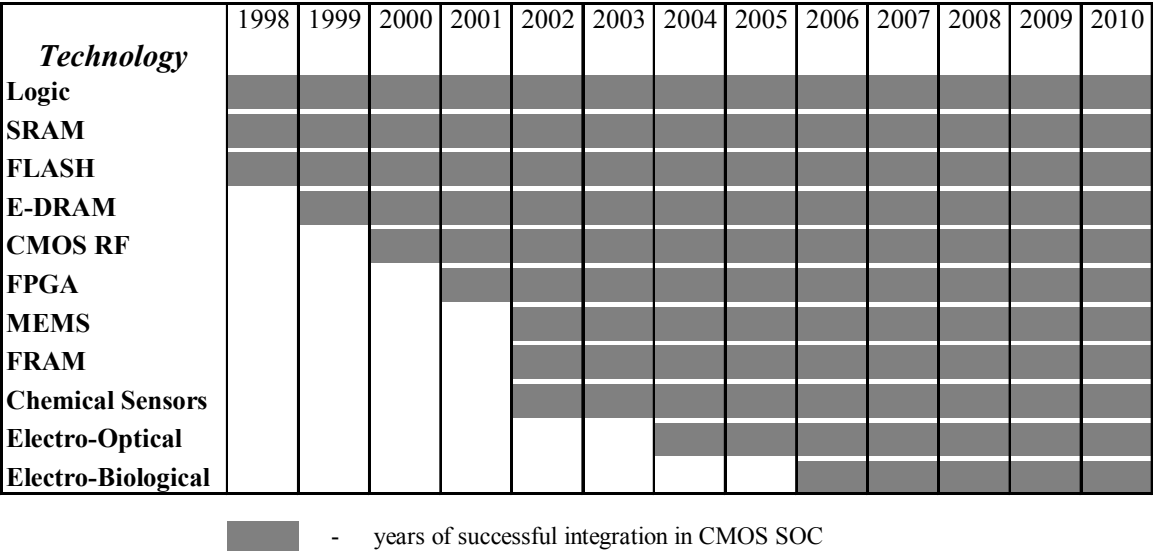


Figure 3: Expected Year for Integration of Technology on SOC Using Standard CMOS Process (Source: SIA, 1999)

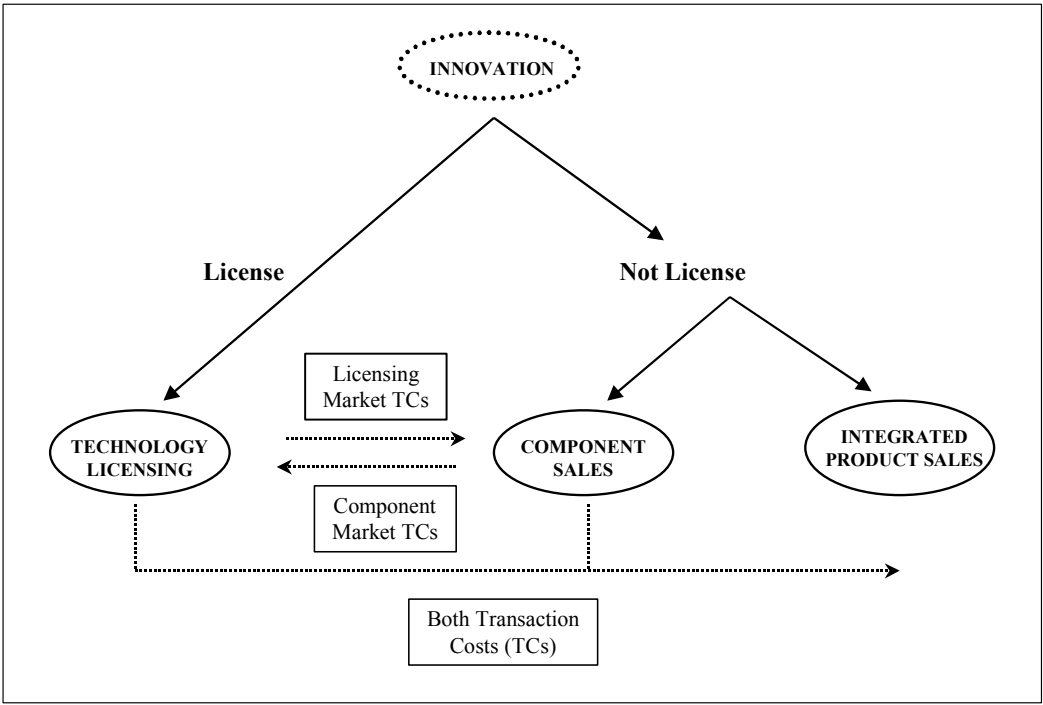


Figure 4: Innovation, Licensing and Integration in Multi-invention Products

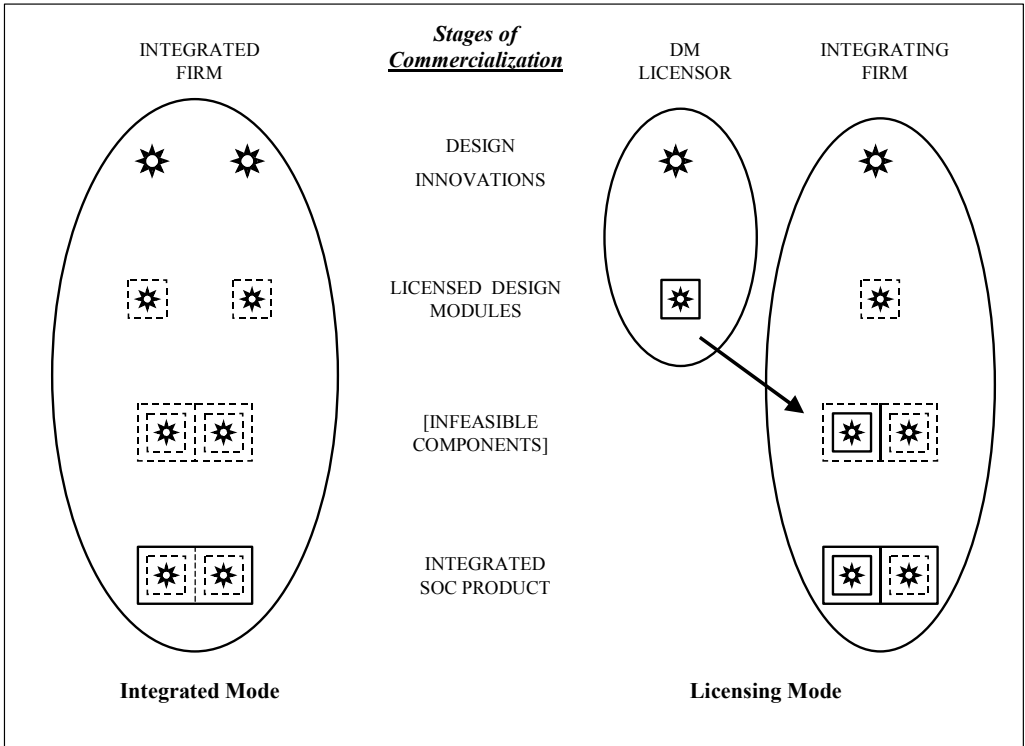


Figure 5: Integrated and Licensing Modes Under SOC Integration

Table 1: System on a Chip Revenues (US\$ Billion)

Year	SOC Revenue	Total Semiconductor Revenue	SOC Share
1997	\$7.60	\$147	5.20%
1998	\$9.40	\$135	6.80%
1999	\$14.00	\$165	8.50%
2000	\$20.80	\$222	9.40%
2001	\$29.10	\$278	10.50%
2002	\$38.00	\$321	11.80%

SOURCE: Dataquest

NOTE: Data for 2000 to 2002 are forecasts as of May 2000

Table 2: Licensing Revenue from Design Modules (US\$ millions)

	1995	1996	1997	1998	1999
"Macro" DMs *	\$16.80	\$48.00	\$96.20	\$140.90	
"All" DMs **				\$330.00	\$442.00

* :Design Modules exceeding 2,500 gates

** : "Pre-designed function[s] to be implemented in a semiconductor device", with no size threshold

SOURCE: Dataquest

Table 3: Worldwide Standard Cell Sales

Rank		Company	Sales (\$ millions)		% change
1997	1998		1997	1998	
1	1	IBM	\$1,532.90	\$1,742.80	13.70%
2	2	Lucent Technologies	\$1,318.30	\$1,554.50	17.90%
4	3	LSI Logic	\$724.00	\$1,241.00	71.40%
3	4	NEC	\$938.10	\$893.60	-4.70%
5	5	Texas Instruments	\$620.20	\$697.30	12.40%
6	6	VLSI Technology	\$595.00	\$601.70	1.10%
7	7	Hitachi	\$495.70	\$401.90	-18.90%
9	8	ST Microelectronics	\$264.70	\$298.80	12.90%
10	9	Alcatel Microelectronics	\$215.00	\$260.00	20.90%
8	10	Toshiba	\$273.20	\$243.80	-10.80%
Top 10 Total:			\$6,977.00	\$7,935.40	13.70%
Total Market:			\$9,112.80	\$9,729.10	6.80%
Total Semiconductor Market:			\$137,203.00	\$125,612.00	-8.40%

SOURCE: CAHNERS IN-STAT GROUP [Bill Arnold, "It Was A Very Bad Year", Electronic Business, May 1999]

Table 4: Relative Costs and Benefits of Licensing and Integrated Modes

Mode	Costs	Source of Cost	Potential Mitigating Factors
LICENSING MODE	Problems with coordinating and solving design problems across firm boundaries	Technological Interconnectedness	Effective "modularizing" public standards
	Inability to identify and contract with IP owners, without opportunism	Diffuse IP Entitlements	Early verifiable IP disclosures; corrective DM trading norms / rules
	Time consuming disputes over DM valuation	Valuation Difficulties	Valuation intermediaries; well defined IP rights; trading rules and arbitration
	Inability to quantify and detect breach of contract	Monitoring and Metering Problems	Technological monitoring solutions; standardized performance metrics; trading rules
	Slow-moving industry standards for DMs that compromises performance	Bureaucratic and Political Public Standards Bodies	-- no real solution --
INTEGRATED MODE	Lacks targeted incentives for creating cutting edge innovative DMs	Governance-based Coordination Within the Firm	Lower costs to acquiring and integrating innovative start-ups
	Poor experimentation with alternative innovative solutions to design problems	Governance-related Intra-firm Cognitive Limitations	
	Less likely to incorporate (or license out) best-in-class DMs	Internal Political Opposition	-- no real solution --

N.B. This table appears to document only costs and no benefits, but the benefits of the licensing mode show up as costs of the integrated mode, and vice versa.